



Khulna University of Engineering & Technology, Khulna

Department of Electronics and Communication Engineering

Established 1967 | Khulna, Bangladesh

Course Code: ECE 2104

Course Title: Digital Electronics and Logic Circuits Laboratory

OPEN ENDED LAB

Traffic Light Controller

GROUP - B-12

ROLL - 2309056 & 2309057

BATCH - 2k23

AIM

To design and implement a digital **traffic light controller system** using fundamental sequential and combinational logic components, simulating a real-world three-phase traffic signal (Red → Green → Yellow) with controlled timing.

PROBLEM STATEMENT

Design a traffic light controller system using digital logic circuits. The system shall cycle through three states — Red, Green, and Yellow — using a 555 Timer as the clock source and D flip-flops (7474) to implement the state machine. The timing interval for each light phase shall be adjustable via a 100 kΩ potentiometer. LEDs of appropriate colours (Red, Green, Yellow) act as the traffic lights. Logic gates (7408 AND) provide the necessary combinational control logic between flip-flop outputs and the LED drivers.

APPARATUS / COMPONENTS REQUIRED

#	Component Name	Specification / Part No.	Qty	Purpose
1	D Flip-Flop IC	7474 (Dual D-type)	2	State storage / sequencing
2	AND Gate IC	7408 (Quad 2-input)	1	Combinational output logic
3	Red LED	5mm, 2V / 20mA	1	Traffic light — Stop
4	Green LED	5mm, 2.2V / 20mA	1	Traffic light — Go
5	Yellow LED	5mm, 2.1V / 20mA	1	Traffic light — Caution
6	555 Timer IC	NE555 / LM555	1	Astable clock pulse generation
7	Potentiometer	100 kOhm	1	Adjustable timing (R1 of 555)
8	Resistor	10 kOhm, 1/4W	3	LED current limiting
9	Capacitor (Electrolytic)	47 uF, 16V	1	Timing capacitor for 555
10	Breadboard & Wires	Standard	1 set	Circuit construction
11	DC Power Supply	+5V regulated	1	Circuit power

MAIN COMPONENTS

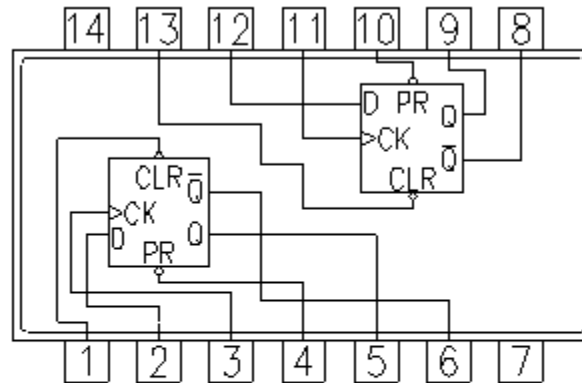
The main components employed in the lab solution are described below along with their pin diagrams and functional descriptions.

1. D Flip-Flop — IC 7474

The 7474 IC is a Dual D-type Positive-Edge-Triggered Flip-Flop with preset (PR) and clear (CLR) inputs. On each rising edge of the clock signal, the output Q follows the data input D.

In this project, two 7474 ICs (providing up to four flip-flops) are used to build a 2-bit ring counter that cycles through three traffic light states: Red (00), Green (01), Yellow (10). The PR and CLR are active-LOW asynchronous override inputs.

Key Specs: Supply: 5V (TTL); Max clock: 25 MHz; Propagation delay: ~14 ns.



7474
Dual D Flip-Flop
with Preset and Clear

Fig 1. 7474 Dual D Flip-Flop — Pin Diagram

2. AND Gate — IC 7408

The 7408 IC is a Quad 2-Input AND Gate containing four independent AND gates. Its Boolean expression is $Y = A \cdot B$ — the output is HIGH only when both inputs are HIGH.

In this project, the 7408 gates combine flip-flop state outputs with the clock enable or direct LED drive signals, ensuring that only the correct LED illuminates for the current traffic state. The gate also prevents conflicting states from simultaneously asserting multiple LEDs.

Key Specs: Supply: 4.75–5.25V; Fan-out: 10 (TTL); Propagation delay: ~11 ns.

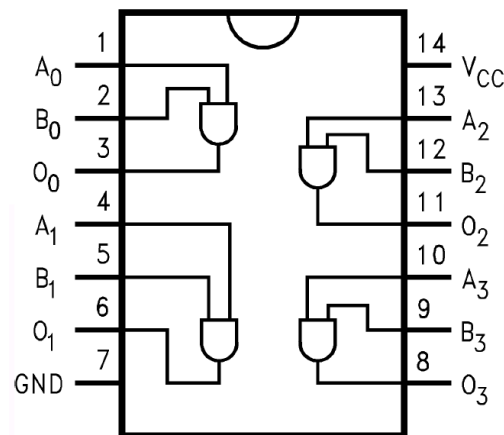


Fig 2. 7408 Quad 2-Input AND Gate — Pin Diagram

3. LEDs — Red, Green, Yellow (5 mm)

Light Emitting Diodes (LEDs) are semiconductor devices that emit light when forward-biased. Each LED has an Anode (+) and a Cathode (–) terminal. The LED emits its characteristic colour depending on the semiconductor material.

In this project, three LEDs represent the three traffic signal phases: Red (Stop), Green (Go), and Yellow (Caution). They are connected via 10 kΩ current-limiting resistors to the flip-flop outputs.

Forward Voltage: Red ≈ 2.0V; Green ≈ 2.2V; Yellow ≈ 2.1V. Forward current: 10–20 mA.

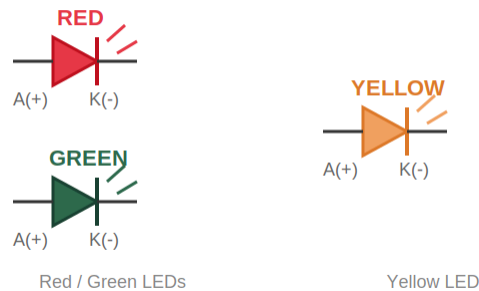


Fig 3. LED Schematic Symbols — Red, Green, Yellow

4. 555 Timer IC — NE555 / LM555

The 555 Timer IC is a versatile integrated circuit widely used for timer, pulse generation, and oscillator functions. It can operate in three modes: Astable (free-running oscillator), Monostable (one-shot), and Bistable.

In this project the 555 is configured in Astable mode — acting as a free-running clock oscillator. The output square wave clocks the D flip-flops. The frequency is controlled by the 100 kΩ potentiometer and 47 μF capacitor, allowing timing adjustment per traffic phase.

Formula: $f = 1.44 / ((R_a + 2R_b) \times C)$ | Supply: 5–15V; Output current: up to 200 mA.

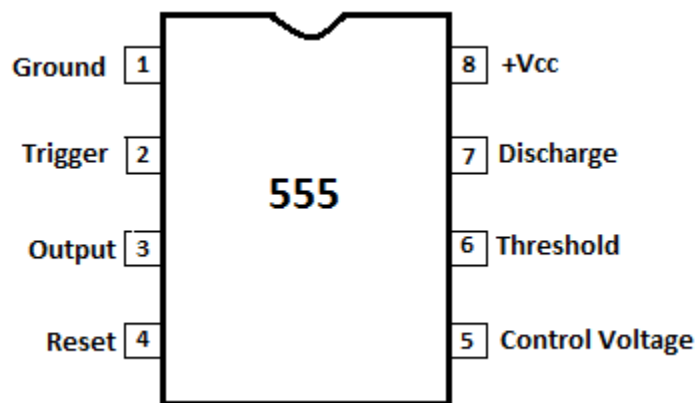


Fig 4. NE555 Timer IC — Pin Diagram

5. Potentiometer — 100 k Ω

A **potentiometer** is a three-terminal variable resistor with a sliding or rotating wiper contact. By adjusting the wiper position, the resistance between the wiper (pin 2) and either end terminal can be varied from 0 Ω to the full 100 k Ω .

In this project, the 100 k Ω potentiometer is used as R_a in the 555 astable circuit, allowing the user to vary the clock frequency — and thus the duration each traffic light remains ON — without modifying the circuit.

Spec: Resistance range: 0 – 100 k Ω ; Type: Rotary carbon track; Power rating: 0.25W.

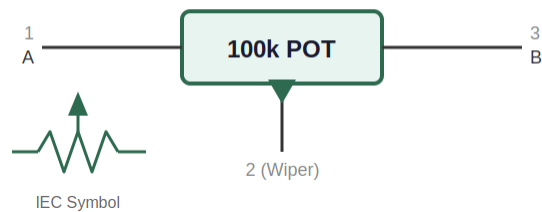


Fig 5. 100 k Ω Potentiometer — Diagram & Symbol

6. Resistor — 10 k Ω , $\frac{1}{4}$ W

A **resistor** is a passive two-terminal electrical component that limits or regulates the flow of current in a circuit.

In this project, 10 k Ω resistors serve as LED current-limiting resistors. Given $V_{cc} = 5V$ and $V_{LED} \approx 2V$: $I = (5 - 2) / 10,000 = 0.3$ mA, safely within LED limits while producing visible illumination. Additional 10 k Ω resistors may be used as pull-ups on flip-flop inputs.

Colour Code: Brown – Black – Orange – Gold (10 k $\Omega \pm 5\%$). Power: $\frac{1}{4}$ W carbon film.

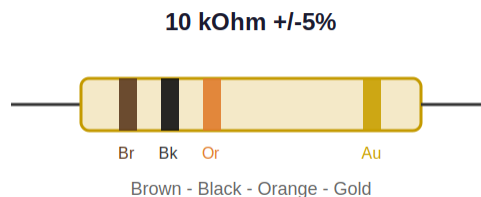


Fig 6. 10 k Ω Resistor — ANSI Symbol with Colour Bands

7. Capacitor — 47 μ F, Electrolytic

An **electrolytic capacitor** is a polarised capacitor that uses a liquid electrolyte as one of its plates. It offers high capacitance in a compact package and must be connected with correct polarity (+ to positive voltage, – to ground).

In this project the 47 μ F capacitor is the timing capacitor (C) in the 555 astable circuit. Together with the 100 k Ω potentiometer, it determines the oscillation frequency: $T \approx 0.693 \times (R_a + 2R_b) \times C$. A larger capacitance results in a slower clock, giving longer light phases.

Spec: Capacitance: 47 μ F; Voltage rating: $\geq 16V$; Type: Aluminium electrolytic; Polarised.

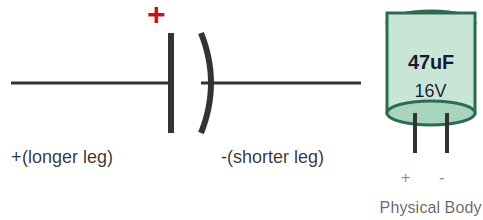


Fig 7. 47 μ F Electrolytic Capacitor — Symbol & Physical Body

CIRCUIT DIAGRAM

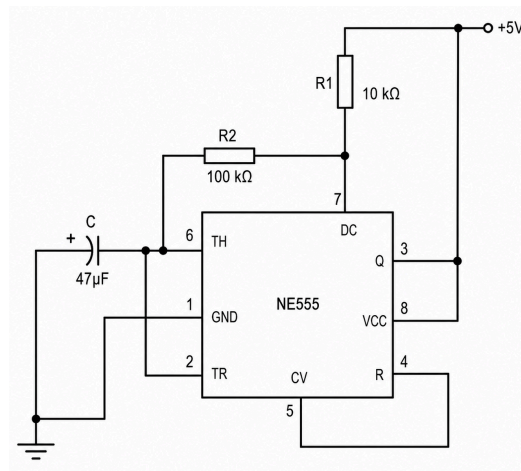


Fig 8. 555 timer circuit to generate clock pulse

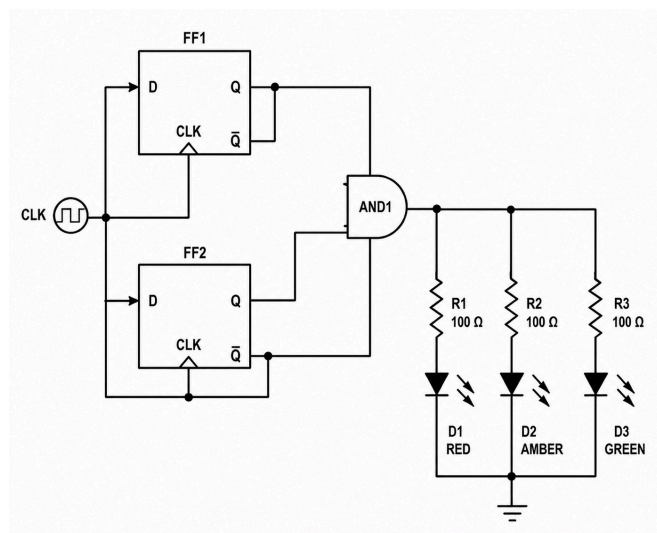


Fig 9. Circuit diagram for Trafficlight system using d flip flops

WORKING PRINCIPLE OF TRAFFIC LIGHT CONTROL SYSTEM USING D FLIP-FLOP AND 555 TIMER

The traffic light control system operates by generating timed sequential signals using a combination of a 555 timer in astable mode and D flip-flops.

555 Timer (Astable Multivibrator): The 555 timer is configured as an astable multivibrator, which continuously produces a square wave clock pulse. The frequency (or time period) of this clock signal is determined by the external resistors and capacitor connected to the 555 timer circuit. This clock pulse acts as the timing source for the entire system.

D Flip-Flop State Storage: The output of the 555 timer is fed into the clock input of the D flip-flops. The D flip-flops are used as state storage elements, meaning they hold and shift the current state of the traffic lights (e.g., Red → Green → Yellow → Red).

Sequential State Transition: Each clock pulse triggers the flip-flops to change their state according to the logic designed in the circuit. By properly connecting the outputs (Q and $\bar{Q}$) of the flip-flops back to their inputs (D), a sequential state transition is achieved. This creates a repeating pattern that corresponds to the traffic light sequence.

LED Output Control: The outputs of the flip-flops are connected to LEDs, which represent the traffic lights:

- One output controls the Red LED
- Another controls the Green LED
- Another controls the Yellow LED

At each clock pulse:

1. The flip-flop changes state.
2. The active output shifts to the next LED.
3. The LEDs turn ON/OFF accordingly, simulating real traffic signals.

Timing Control: The duration of each light depends on the clock pulse interval generated by the 555 timer. By adjusting the resistor and capacitor values, the timing of the traffic lights can be controlled (e.g., longer red light, shorter yellow light).

Thus, the system continuously cycles through predefined states, creating an **automated traffic light control mechanism**.
